1 $S_n = S_{n-1} + 2S_{n-2}$

a) $r^n = r^{n-1} + 2r^{n-2}$ $\quad r^2 = r + 2$ $\quad r^2 - r - 2 = 0$ $\quad (r-2)(r+1) = 0$ $\quad r = \{2, -1\}$

$a_n^{(h)} = \alpha_1 (2)^n + \alpha_2 (-1)^n$ $\quad a_0 = \alpha_1 + \alpha_2 = 3$ $\quad a_1 = 2\alpha_1 - \alpha_2 = -3$

$3\alpha_1 = 0 \rightarrow \alpha_1 = 0, \alpha_2 = 3$ $\quad \boxed{S_n = 3 \cdot (-1)^n}$

b) $S_n = 2S_{n-1} - S_{n-2}$ $\quad r^n = 2r^{n-1} - r^{n-2}$ $\quad r^2 = 2r - 1$ $\quad r^2 - 2r + 1 = 0 = (r-1)^2$

$r = 1$

$S_n = \alpha_1 (1)^n + \alpha_2 \cdot n \cdot (1)^n = \alpha_1 + n \cdot \alpha_2$ $\quad S_0 = \alpha_1 = 3$ $\quad S_1 = \alpha_1 + \alpha_2 = 5$

$\hookrightarrow \alpha_2 = 2$

$S_2 = 10 - 3 = 7$

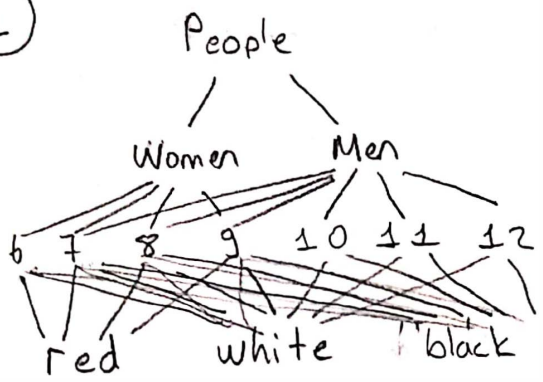
$S_3 = 14 - 5 = 9$

$\boxed{S_n = 3 + n \cdot 2}$

c) $S_{n+1}^2 = 5 S_n^2$ $\quad r^{n+1} \cdot r^{n+1} = 5 \cdot r^n \cdot r^n$ $\quad r^2 = 5$ $\quad r = \sqrt{5}$

$S_n = \alpha_1 (\sqrt{5})^n$ $\quad S_0 = \alpha_1 = 2$ $\quad \boxed{S_n = 2(\sqrt{5})^n}$ $\quad \boxed{S_6 = 2(\sqrt{5})^6 = 250}$

(2)



Men: $6 \cdot 2 = 12$ shoes

Women: $4 \cdot 3 = 12$ shoes

Men + Women: $3 \cdot 2 = 6$ shoes

Total = $(12+12) - 6 =$ **18 different shoes**

(3) a)

	deg^-	deg^+	out - in
a	1	1	0
b	3	4	1
c	1	1	0
d	2	2	0
e	3	2	-1

no euler circuit, it has an euler path

b - a - d - e - c - b - e - b - d - b - e

b)

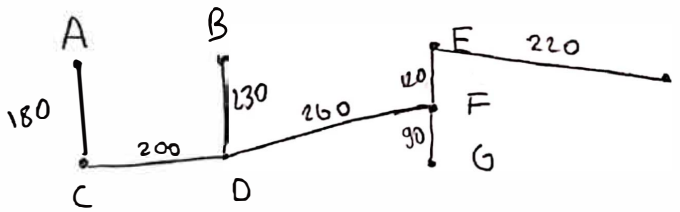
	u_1	u_2	u_3	u_4	u_5
u_1	0	1	0	1	1
u_2	1	0	1	1	1
u_3	0	1	0	1	0
u_4	1	1	1	0	1
u_5	1	1	0	1	0

$deg(u_1)=3$ $deg(u_3)=4$
$deg(u_2)=4$ $deg(u_4)=3$
$deg(u_3)=2$ $deg(u_5)=4$
$deg(u_4)=4$
$deg(u_5)=3$
$deg(v_1)=3$
$deg(v_2)=2$

	v_1	v_3	v_2	v_5	v_4
v_1	0	1	0	1	1
v_3	1	0	1	1	1
v_2	0	1	0	1	0
v_5	1	1	1	0	1
v_4	1	1	0	1	0

They are isomorphic

c)



Total weight = 180+200+230+260+210+220
350+430+430 = 1300

(4) Base case n=5

$2^5 > 5^2$ $32 > 25$ (true)

$2^6 > 6^2$ $64 > 36$ (true)

if it is true for $n>4$ then true for $n+1>4$

(5) It is reflexive ($M_{ij}=1$ when $i=j$)

It is symmetric $M_R^T = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} = M_R$

It is not antisymmetric because $M_{12}=M_{21}$, $M_{13}=M_{31}$, $M_{32}=M_{23}$